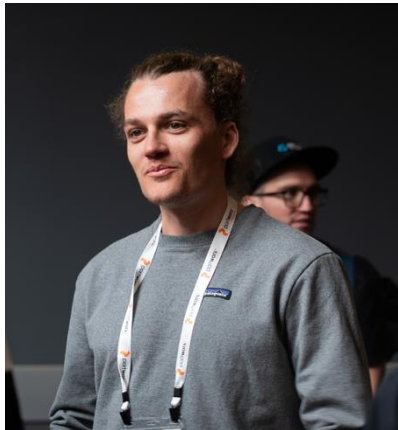


The Accidental Platform Team: Kubernetes Operators at Swisscom

17. September 2026, Lea Brühwiler, Fabian Schulz





Fabian Schulz

DevOps Engineer

fabian.schulz1@swisscom.com



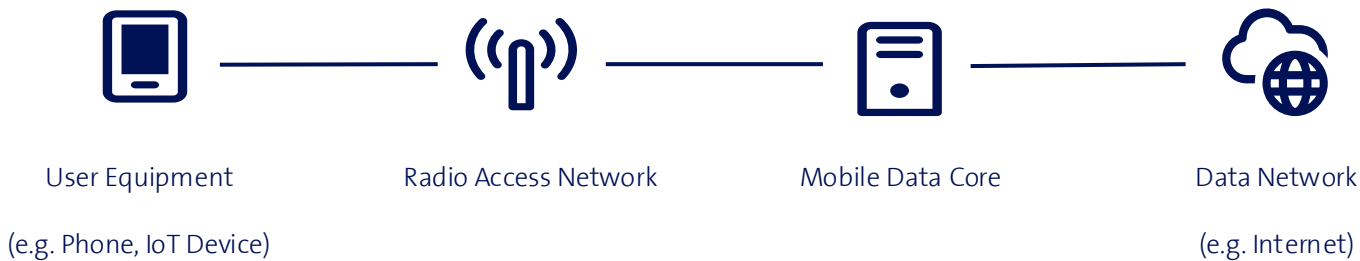
Lea Brühwiler

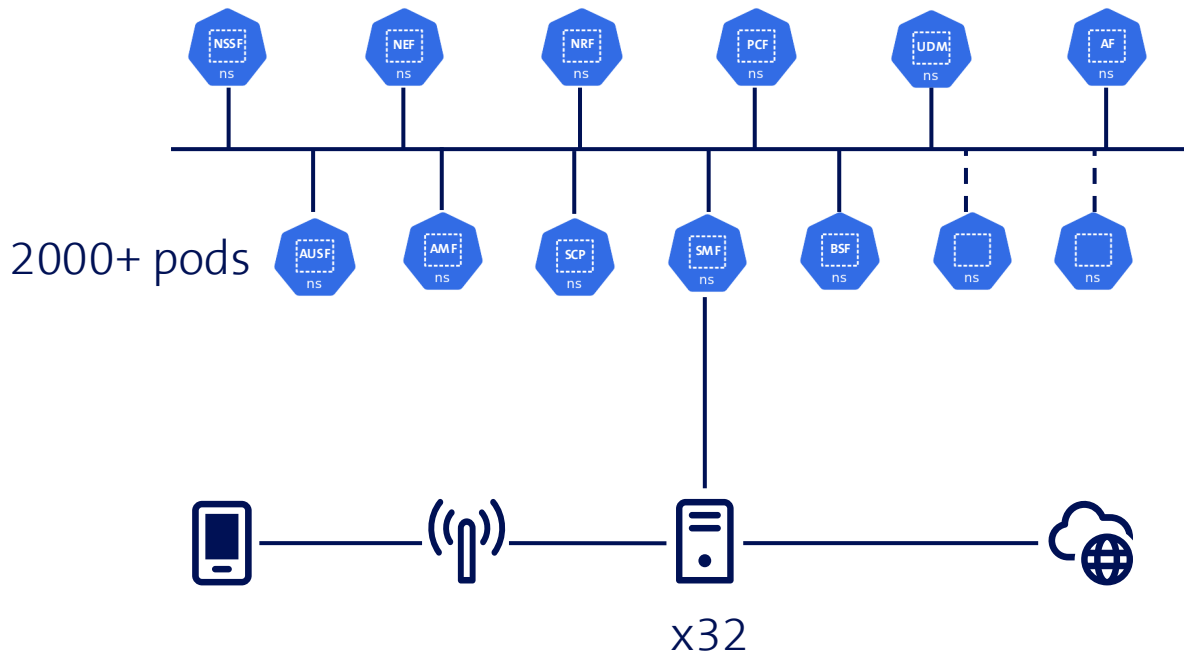
DevOps Engineer

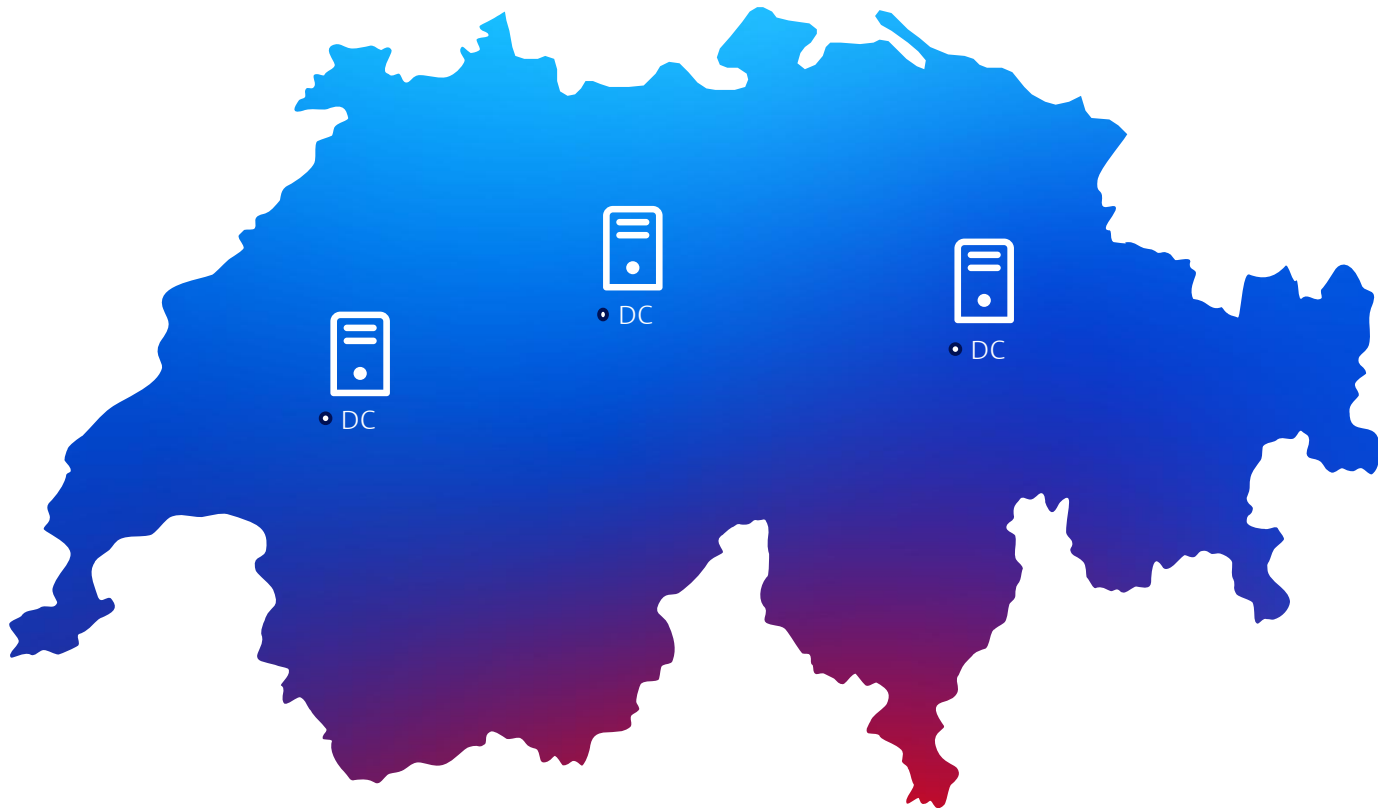
lea.bruehwiler@swisscom.com

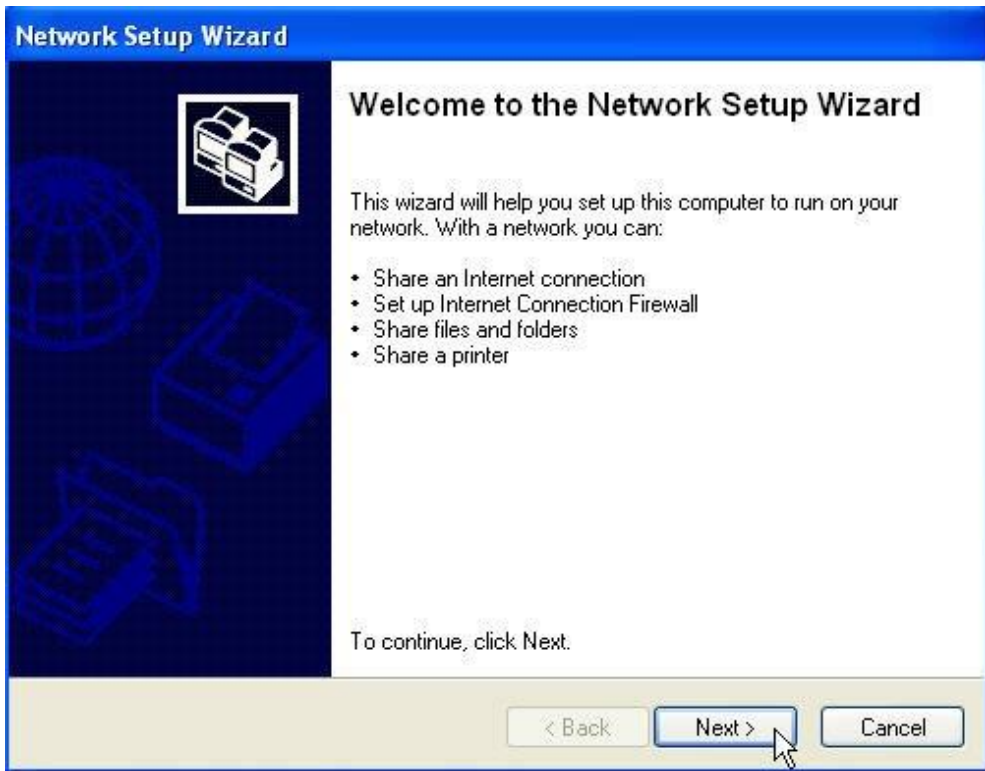


swisscom

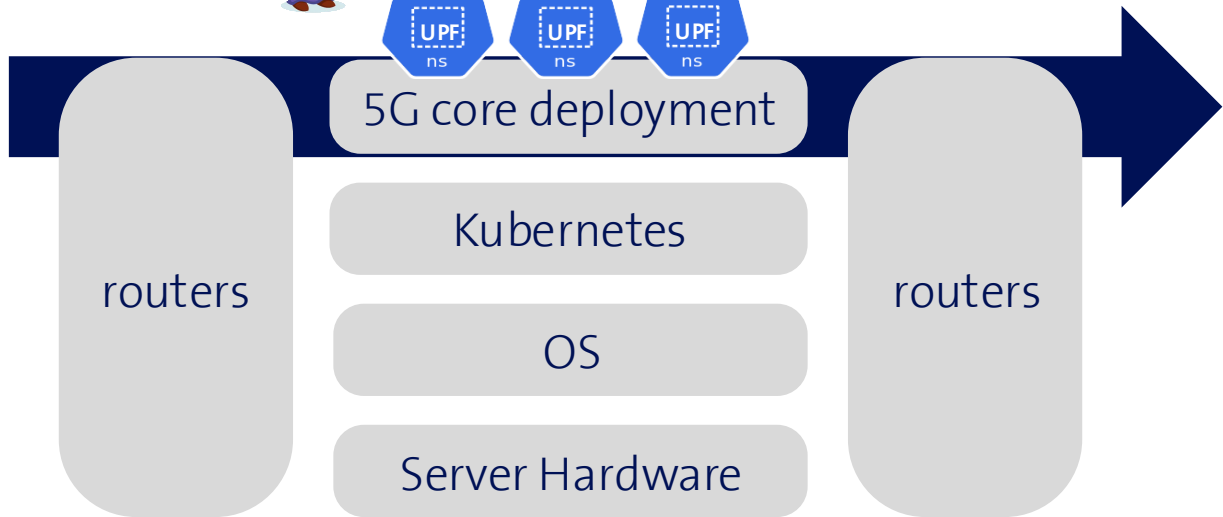














Create IP assignments
2 days



Collect IP in IPAM
2 days



Populate inventory
1 day

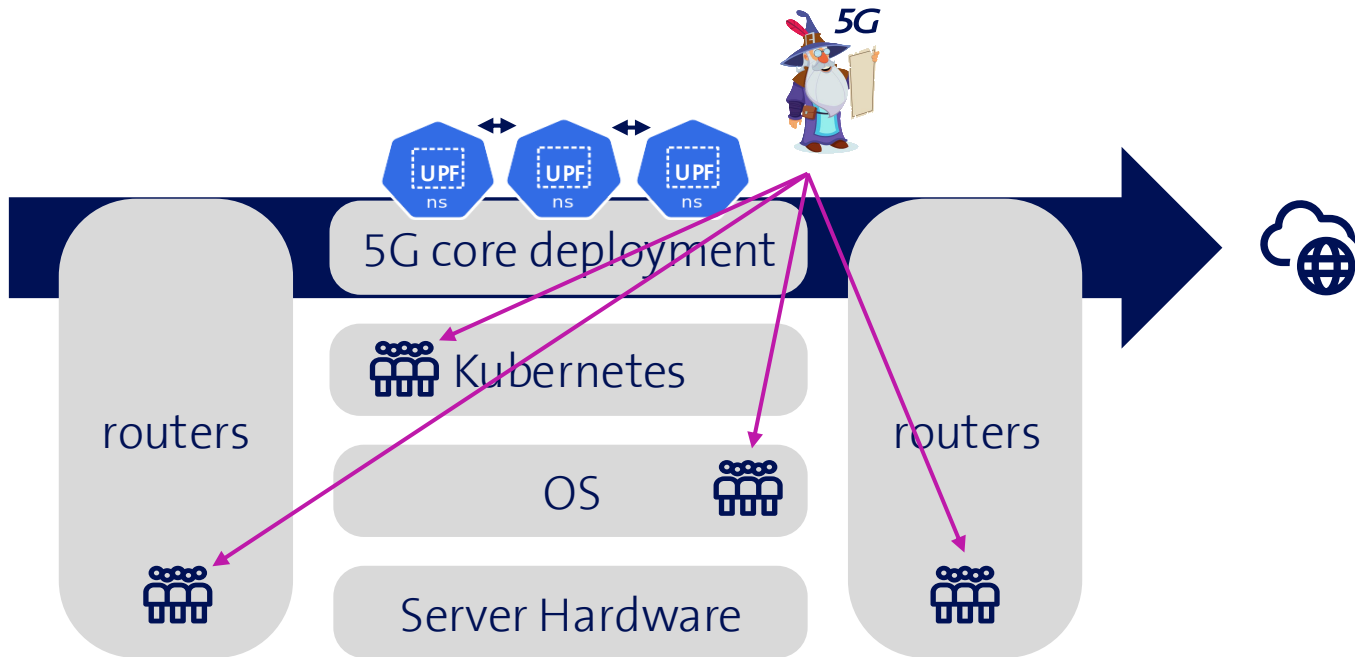


Create change requests
5-10 days



Integrate, test and observe
5 - 10 days







Easy, right?





Pain Points When Deploying a 5G Core Network



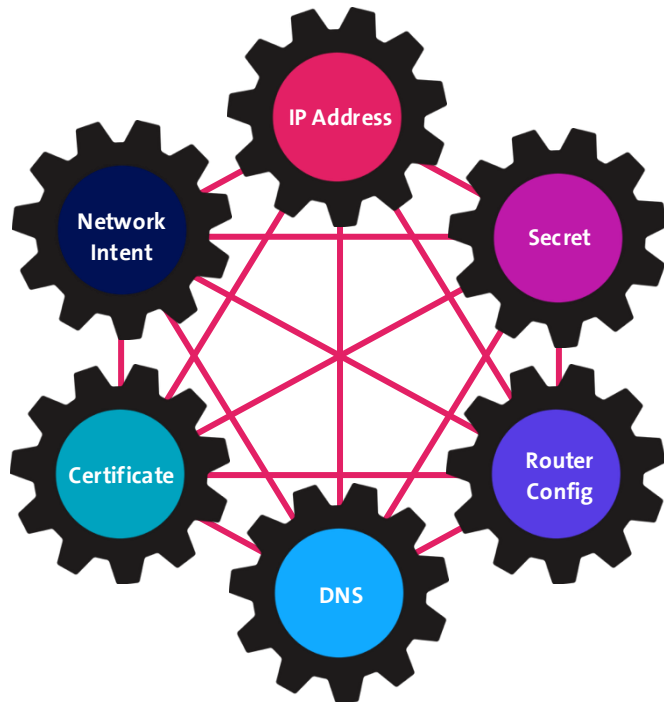
Cross-domain Knowledge

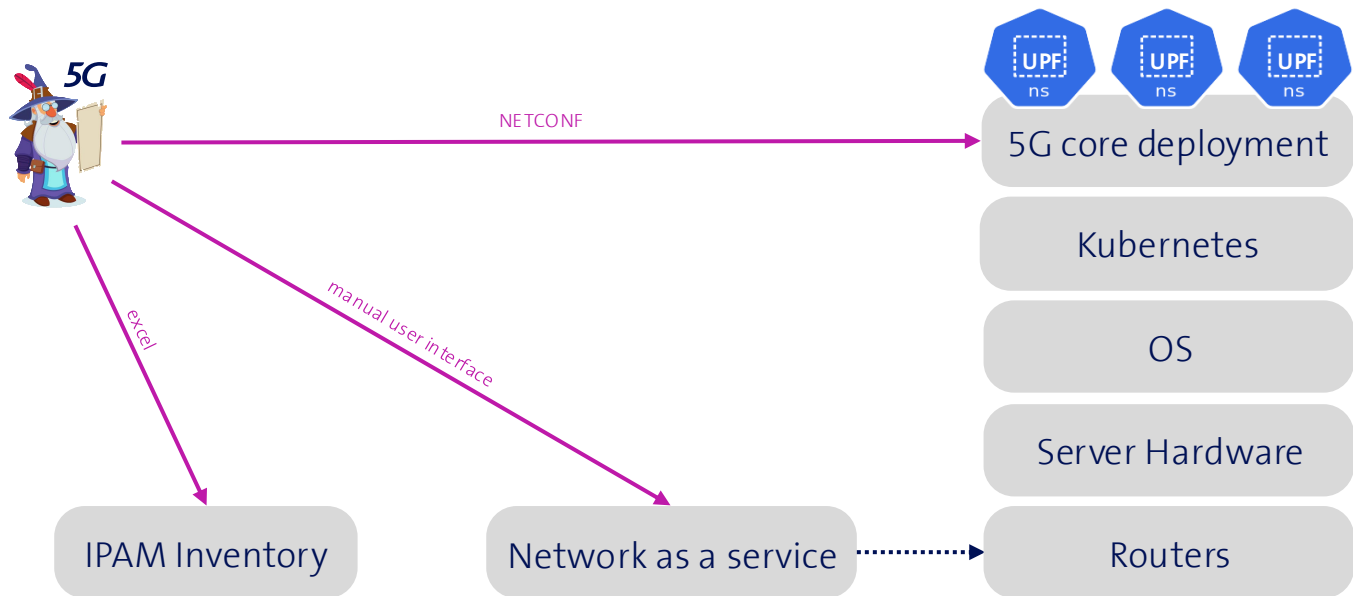


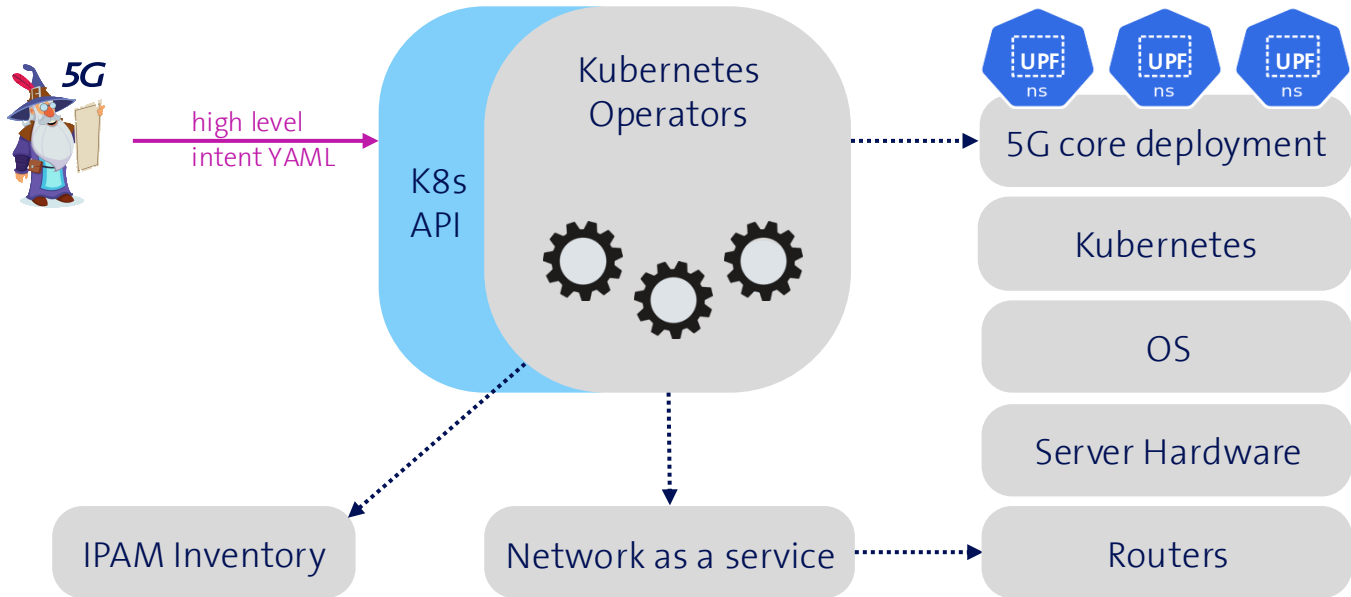
Manual Process



Static Config









**Cross-domain
Knowledge**



**Domain Knowledge
in Operators**



**Manual
Process**



Automation



**Static
Config**



**Constant
Reconciliation**



More details?

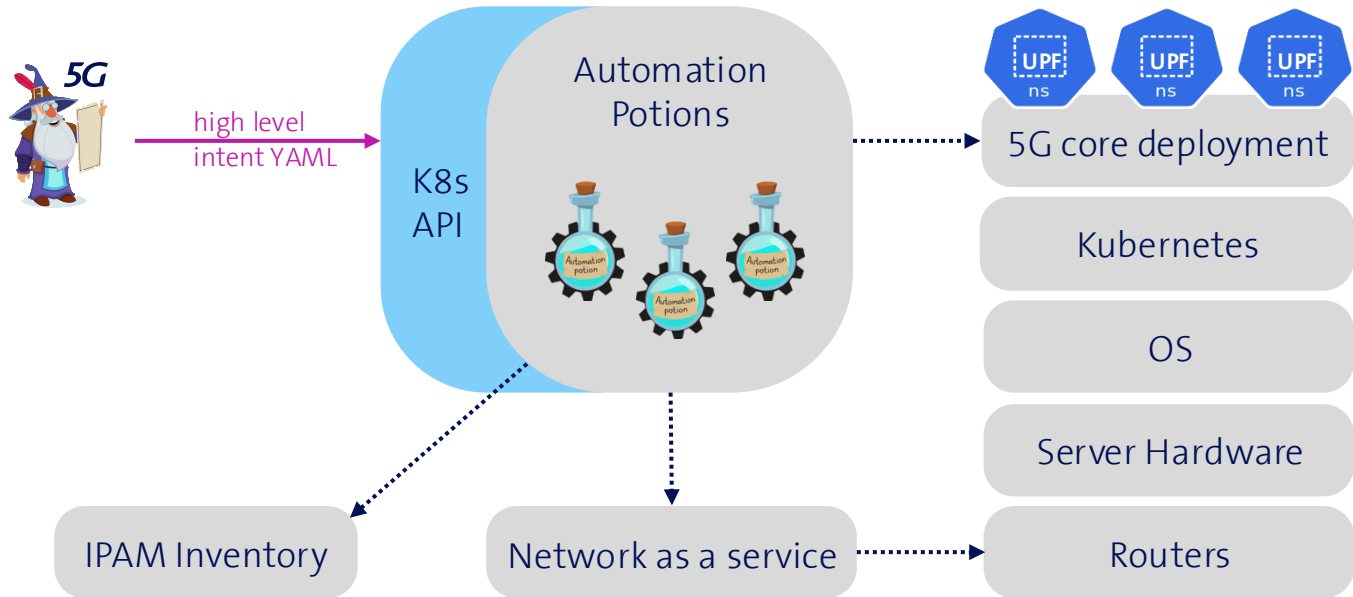
The Accidental Platform Team Swisscom, September 2026





How brave do you feel?







Gradual Approach



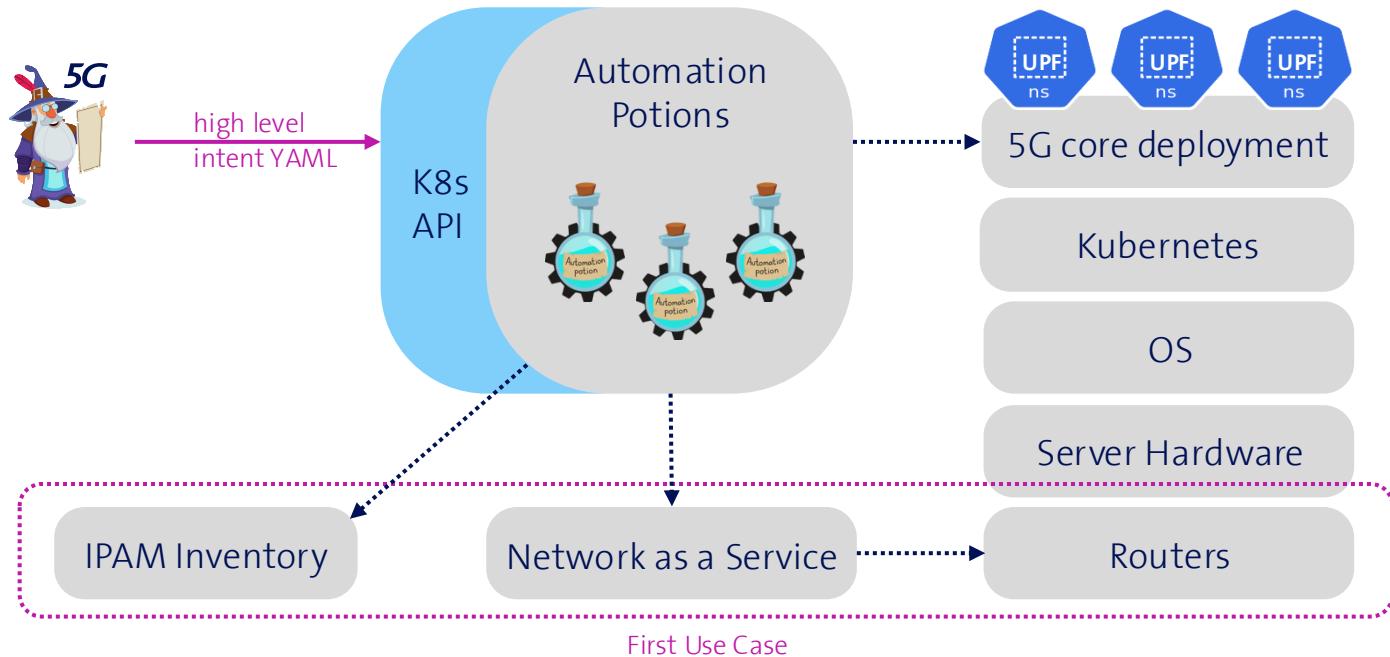
**Select first easy use
case**

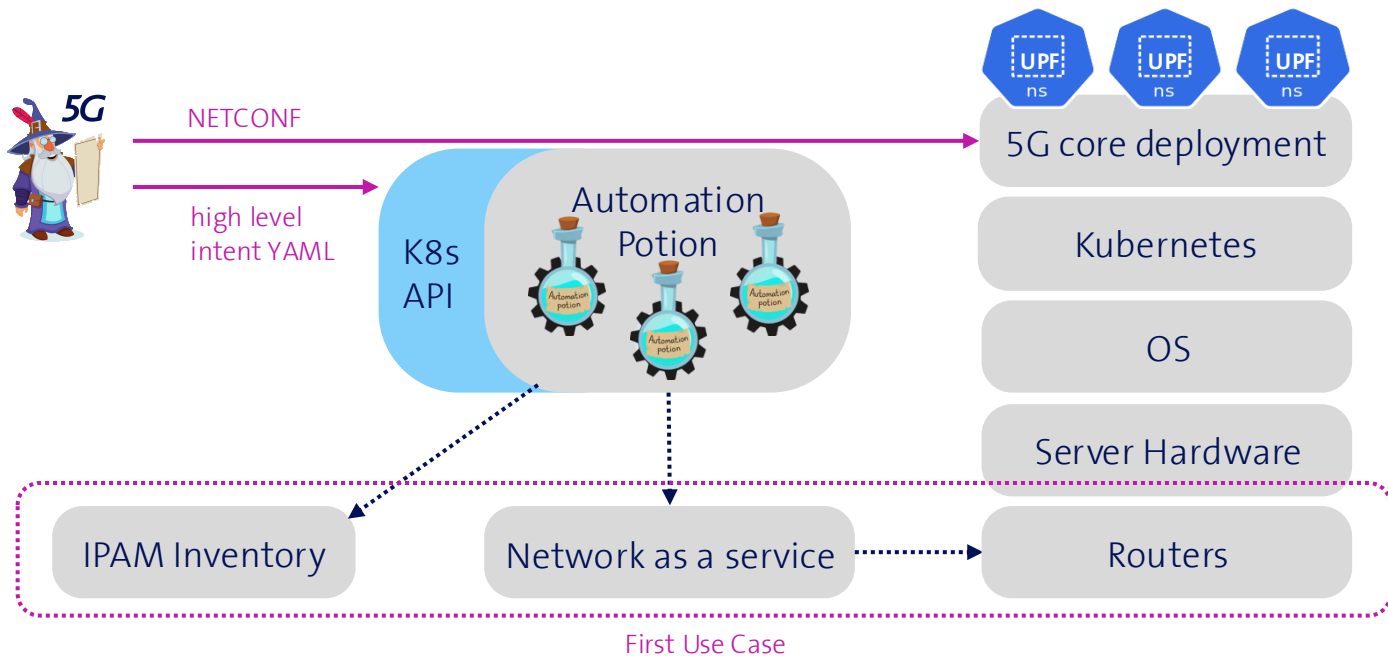


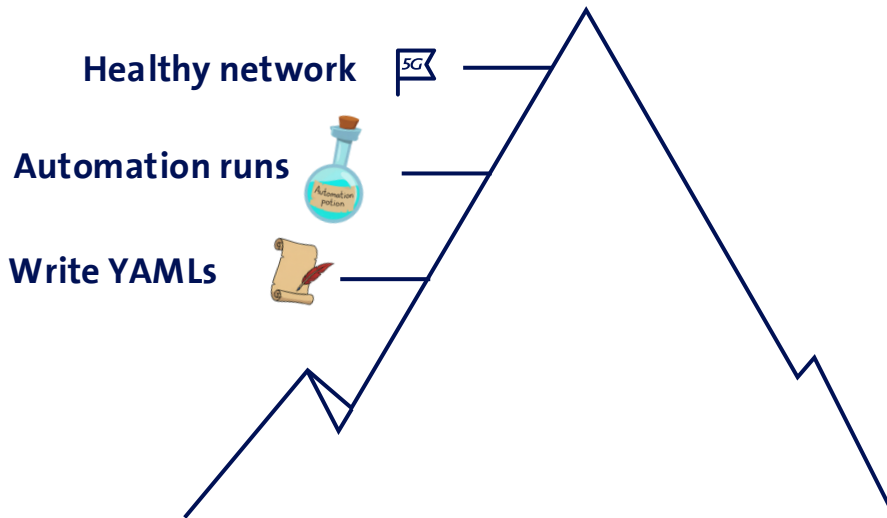
**Ship first prototype
early**



Iterate on feedback







Let's gooooo



We are done, right?



A cartoon illustration of a wizard with a long white beard, wearing a blue robe and a tall blue hat with a red feather. He is holding a large, rolled-up scroll.

```

    required {
      validation: rule = "self == oldSelf", message = "Field 'ipv6' is immutable"
      `json:"ipv6,omitempty"`
    }
  }
}

// Configuring IPv6 IPs, and the selectors used

// Required
required {
  validation: rule = "self == oldSelf", message = "Field 'ipv4' is immutable"
  `json:"ipv4,omitempty"`
}

// IPv6 selection criteria.
enable = (self.enable == false || (has(self.parentPrefixSelector) && has(self.prefixLength)))

// Required
required {
  // In NetBox Operator. Note that the "family" field will be added by the Controller
  // family (IPv4/IPv6)
  validation: rule = "self == oldSelf", message = "Field 'parentPrefixSelector' is immutable"
  ,string `json:"parentPrefixSelector,omitempty"`
}

// For the IPv6 prefix. Be aware that this is an integer which will be converted to
// NetBox Operator

minimum = 8
maximum = 128
validation: rule = "self == oldSelf", message = "Field 'prefixLength' is immutable"
`json:"prefixLength,omitempty"`

```



Maybe not quite:

- We don't have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.
- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!
- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?
- I don't understand Kubernetes.
- I don't even know how to start using it.
- I just did it myself using a Gitlab pipeline, it's easier that way.





- We don't have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.



Gradual Adoption





How do we get our solution tested?

Focus on early adopters



high level
intent YAML

K8s
API

Kubernetes
Operators



Gradually adopt
more use cases



5G core deployment

Kubernetes

OS

Server Hardware

IPAM Inventory

Network as a service

Routers



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Gradual Adoption



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Graphical UI





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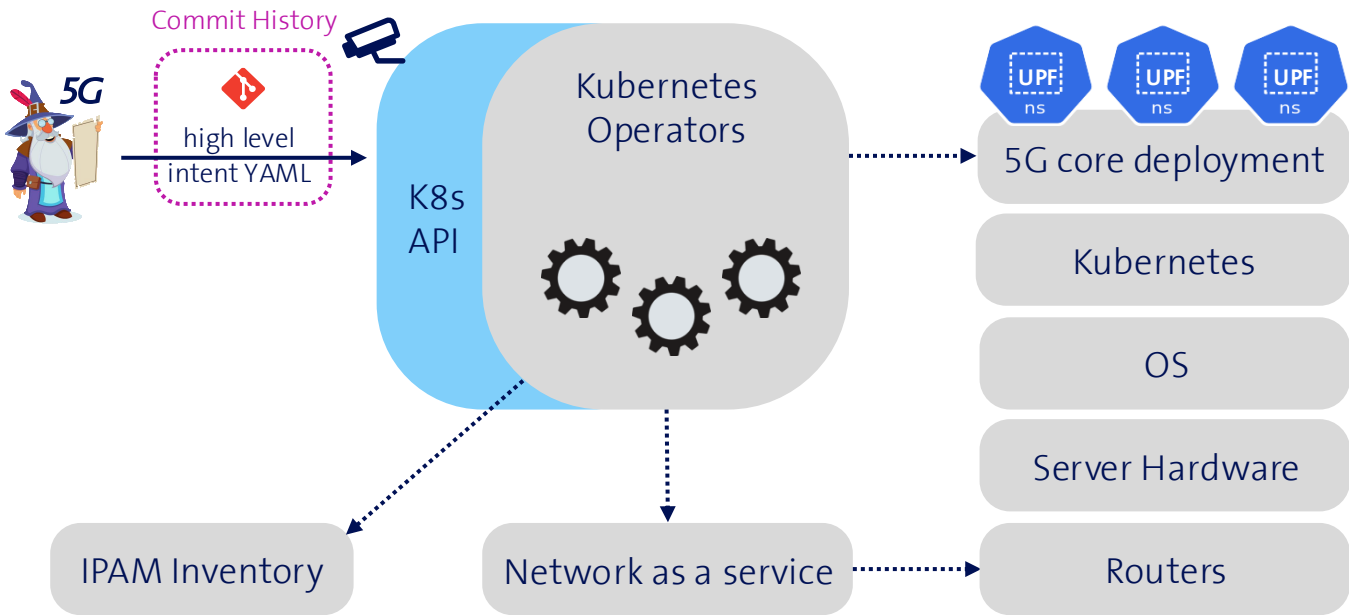


**Observability
and Traceability**





How do we know what happened?



IP Range Claims

134

27

Network Instances

6

Status

Static Routes

Status

Ready

Ready



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Gradual Adoption



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Knowledge Sharing





Establish Trust with Users

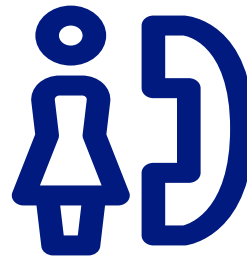
Design / Feedback Sessions



Focused Communication Channels

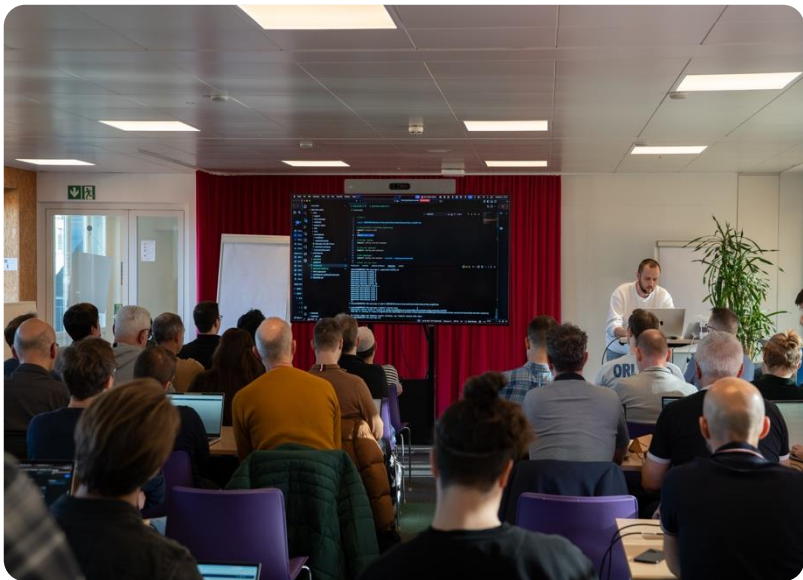


Regular Check-ins





Knowledge-Sharing Initiatives





➤ We don't have time to test your stuff.

➤ You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.



Gradual Adoption



➤ How can I see which changes will be applied to the cluster?

➤ We had a UI which was more intuitive!



Graphical UI



➤ How can I be sure it worked?

➤ I think we were faster the old way.

➤ How do we know where this change is coming from?



Observability and Traceability



➤ I don't understand Kubernetes.

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Knowledge Sharing



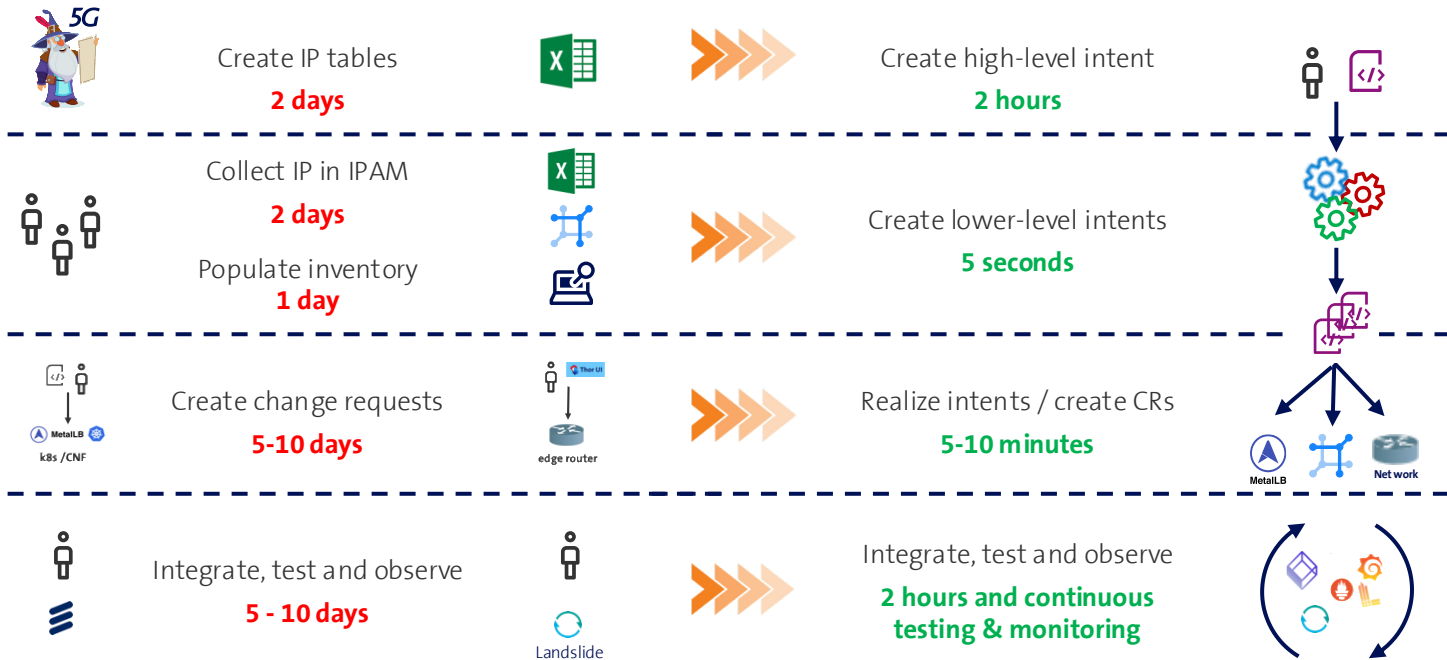


The Tale of the Accidental Platform Team





Does it actually work?





And how brave do you feel now?





Thanks to colleagues
for what we've
achieved **together** in
the past **1.5 years**.

We just created all
endpoints **without a
single error!**

I'm convinced that
we've **saved a week
or two of toil**.



Now, it took me **less
than one hour** to
create those
networks!



What does the future hold?

Automation of all steps

What do we do regarding manual steps?

Different types of systems

How about imperative systems?

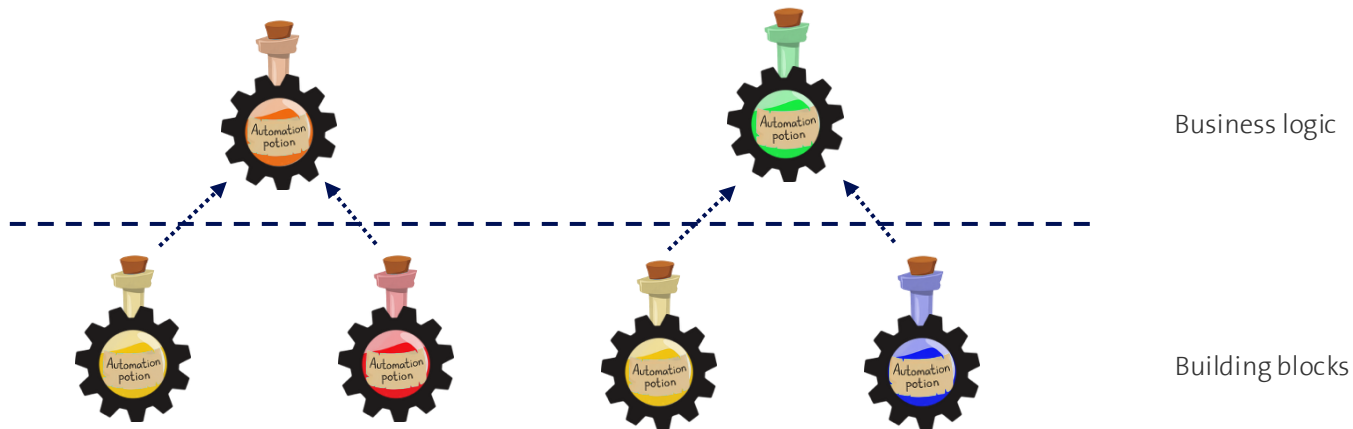
AI

Where can we meaningfully use it?



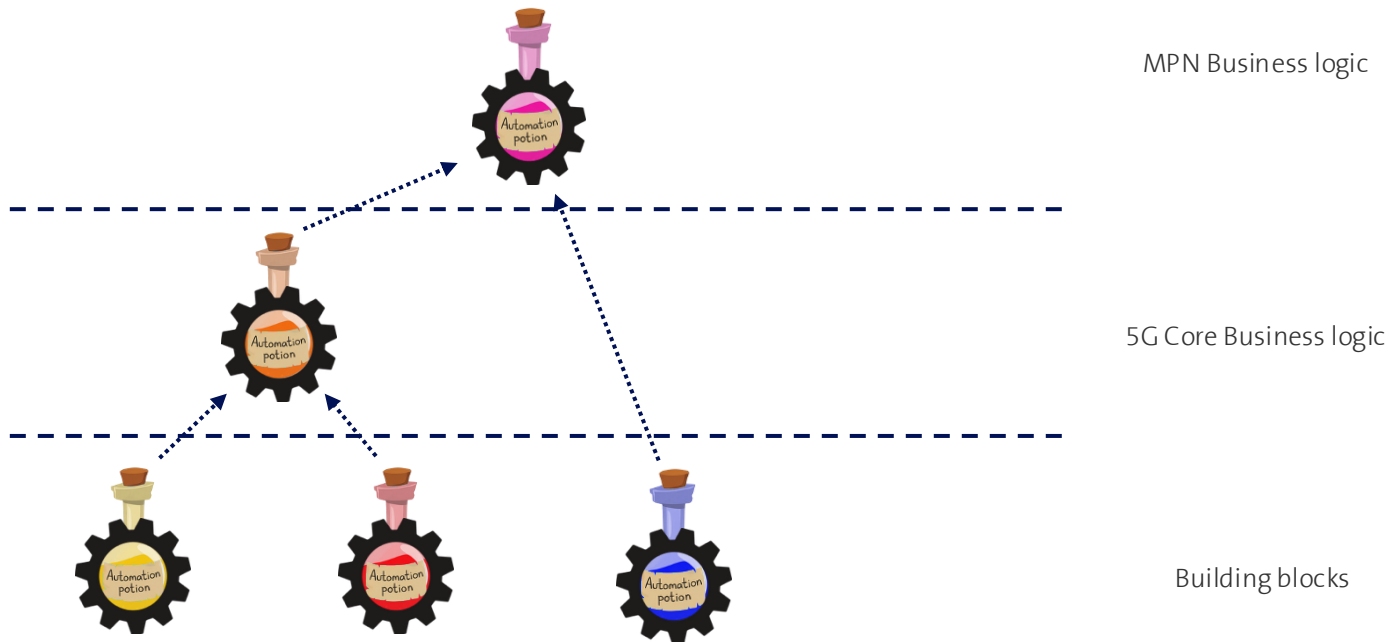


The Operator Stack



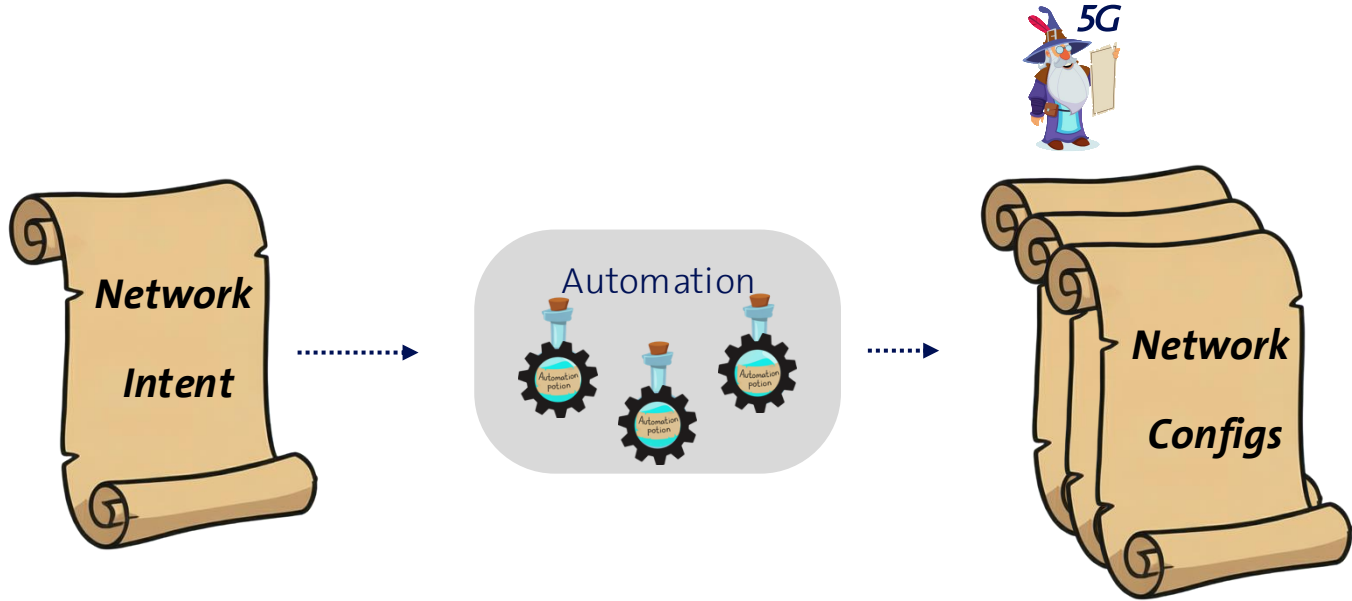


Mobile Private Networks





Custom Operators





Kube Resource Orchestrator -> kro.run

RGD

kind: Resource Graph Definition

spec:

schema:

kind: NetworkConfig

spec:

netName: string

ipv4: net.ip

...

resources:

- id: dnsEntryV4

template:

kind: DnsEndpoint

spec:

cname: 5g- $\{\text{schema.spec.netName}\}$

...



CRD

kind: CRD

spec:

name:

kind: NetworkConfig

schema:

netName: string

ipv4: net.ip



Kubernetes Resource Orchestrator

CRD

```
kind: CRD
spec:
  name:
    kind: NetworkConfig
  schema:
    netName: string
    ipv4: net.ip
```

CR

```
kind: NetworkConfig
spec:
  netName: upf-1
  ipv4: 10.0.12.3
```

RGD



CR

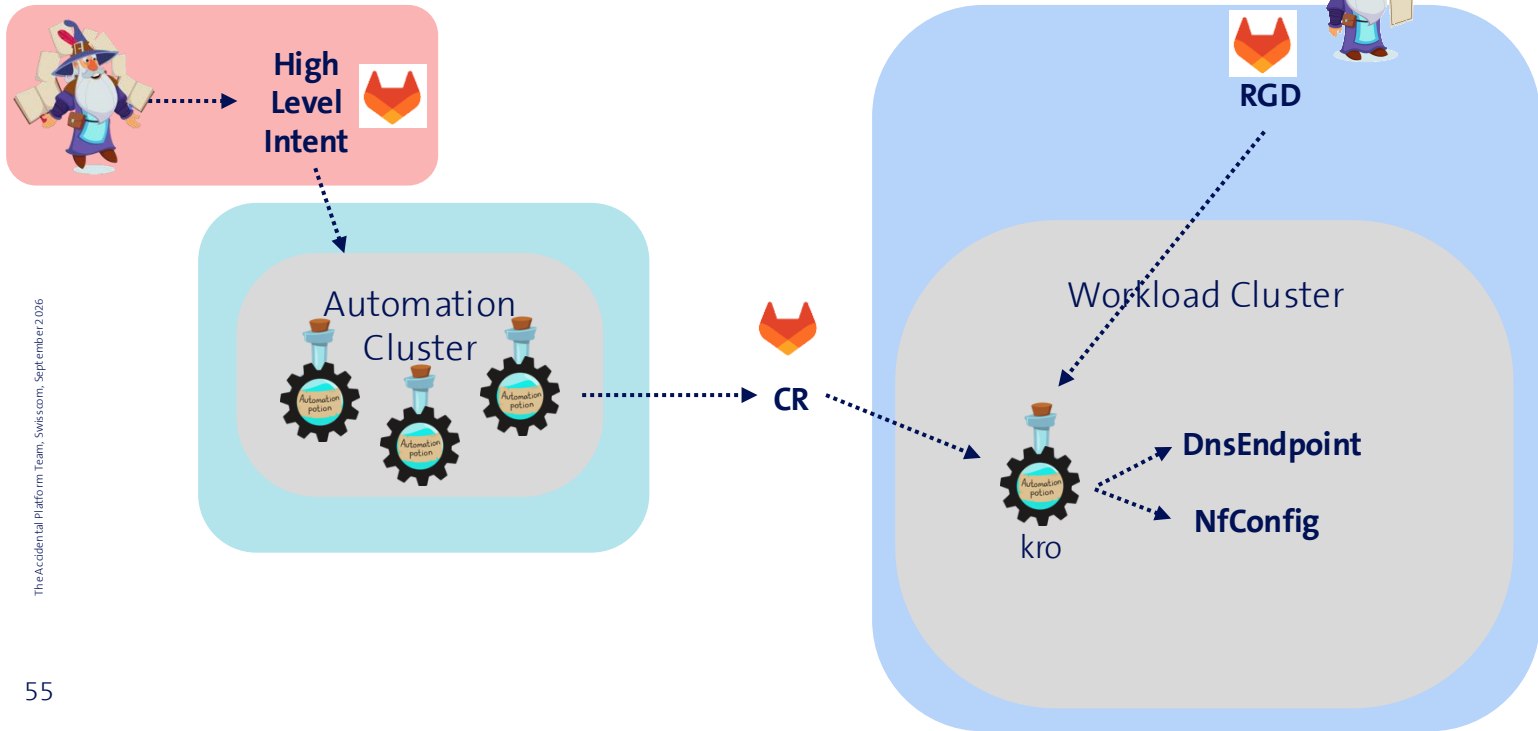
```
kind: DnsEndpoint
spec:
  cName: 5g-upf-1
  destV4: 10.0.12.3
```

CR

```
kind: NfConfig
spec:
  target: upf-1
  route: 10.0.12.3
```



Kubernetes Resource Orchestrator





How to survive as an Accidental Platform Team?





**How We Are Moving from
GitOps to Kubernetes
Resource Model in 5G Core**

KubeCon Europe

2024

**Just Claim It: Simplifying
Network Automation with
NetBox Operator**

Cloud Native Telco Day

2025

**Why Is It So Hard to Run a 5G
Core on Kubernetes—And
What Needs to Change for 6G**

KubeCon + CloudNativeCon
Europe

2026

2024

**Building and Operating a
Highly Reliable Cloud Native
DNS Service With Open-Source
Technologies**

Container Days Conference

2025

**The Cloud-Native Advantage:
Intent-based Network
Automation (SDC)**

Container Days Conference

[illegible]

github.com/swisscom/cloud-native-telco